

Georgios Alexandros Asvestas

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RESEARCH INTERESTS

Geospatial artificial intelligence (GeoAI) for archaeological prospection; multi-sensor UAV remote sensing (LiDAR, multispectral, thermal); data fusion; explainable AI and the faithfulness of attribution methods (SHAP, Grad-CAM++); predictive modelling and spatial statistics for excavation planning; fortified and rural landscapes of the Aegean and Megaris.

AWARDS & DISTINCTIONS

2nd Place, Adaptation to Climate Change Mission Arena - [EU TalentOn 2026](#) Sep 2026

European Commission, EU TalentOn 2026, Brest, France (7–11 September 2026)

- 5000€ awarded to team “Oldtimers” (four-member international team) for *Guardian*, an early-warning system built on the Galileo Emergency Warning Satellite Service (EWSS) that delivers disaster alerts to elderly people and people with disabilities without relying on smartphones or cellular networks

EDUCATION

PhD in Geography (Geoinformatics) 2026 – present

University of the Aegean, Department of Geography, Mytilene

- Thesis: *Advancing the Application of Geospatial Artificial Intelligence to Remote Sensing Data for the Detection and Interpretation of Archaeological Landscapes*.
- Supervisor: Assoc. Prof. Christos Vasilakos

MSc in Geography and Applied Geoinformatics (Geoinformatics) 2023 – 2025

University of the Aegean, Department of Geography, Mytilene

- Overall grade 8.92/10 (Excellent)
- Thesis: *Applied Geoinformatics in Archaeological Research: A Predictive Modelling Framework Combining Multi-Sensor UAV Data* (10/10). Supervisor: Assoc. Prof. Christos Vasilakos

MA in Greek and Eastern Mediterranean Archaeology 2017 – 2018

National and Kapodistrian University of Athens, Department of History and Archaeology

- Overall grade 7.99/10 (ECTS B)
- Thesis: *The Defensive Network of Western Attica* (9.5/10). Supervisor: Dr David Scahill

BA (Ptychio) in Archaeology 2010 – 2017

University of Thessaly, Department of History, Archaeology and Social Anthropology

- Thesis: *Roman Eleusis*. Supervisor: Assoc. Prof. Giannis Lolos

CONFERENCE PRESENTATIONS

Asvestas, G. A., & Vasilakos, C. (2025). Multi-sensor UAV remote sensing workflow for detecting buried archaeological features at Thermi (Lesvos) and its application to Arisvi (Lesvos). *9th Symposium of the Hellenic Society for Archaeometry*, Patras, 5–8 November. Oral presentation. Extended Abstract submission

Asvestas, G. A., & Vasilakos, C. (2025). Integrating UAV multi-sensor data for archaeological detection at Thermi, Lesvos. *Panhellenic Conference of the Hellenic Society for Non-Destructive Testing*, Ormylia Foundation, Chalkidiki, 9–11 May. Oral, Abstract Submission

RESEARCH EXPERIENCE

MSc research: multi-sensor UAV predictive modelling at Thermi and Arisvi, Lesvos 2023 – 2025

Remote Sensing & GIS Research Group, Department of Geography, University of the Aegean

- Developed a non-invasive UAV workflow fusing LiDAR, thermal, multispectral and RGB data to detect buried architecture at Early Bronze Age Thermi, using the legacy excavation plan as ground truth
- Built an interpretable PCA-logistic regression model (AUC 0.964) that outperformed every single-sensor model, and transferred it without refitting to the unexplored hilltop site of Arisvi
- Translated model predictions into a defensible fieldwork plan: a two-stage Getis-Ord Gi* / Local Moran's I analysis narrowed Arisvi to 1,440 priority cells (3,240 m²), a 13-fold reduction in area to investigate; Peer Review process is completed for publication to Journal of Archaeological Science: Reports.

MA research: the defensive network of Western Attica

2017-2018

Department of History and Archaeology, National and Kapodistrian University of Athens

- Synthesis of rural fortifications from the Archaic to the Hellenistic period: catalogue of 42 towers, forts and fortification walls across Parnitha, Skourta, Pateras, Kithaironas, Aigaleo and the Corinthian Gulf coast, with typology and discussion of their military and agricultural functions

PROFESSIONAL EXPERIENCE

Contract Archaeologist

Oct 2022 – present

Ephorate of Antiquities of Lesbos, Hellenic Ministry of Culture

- Systematic and rescue excavations

Contract Archaeologist

Nov 2021 – Sep 2022

Ephorate of Antiquities of Western Attica (Megara–Kineta), Hellenic Ministry of Culture

- Systematic and rescue excavations

Contract Archaeologist

Feb 2020 – Nov 2021

INTRAKAT, under the supervision of the Ephorates of Antiquities of East Attica and of Piraeus and the Islands

- Rescue archaeology on development works

Contract Archaeologist

Oct 2018 – Nov 2019

Ephorate of Antiquities of Western Attica (Megara), Hellenic Ministry of Culture

- Systematic and rescue excavations

TECHNICAL SKILLS

UAV data acquisition: mission planning in DJI Pilot for DJI Mavic 3E (RGB), Matrice 300 RTK with Zenmuse L1 (LiDAR) and H20T (thermal), and Phantom 4 Multispectral; RTK-GNSS / HEPOS positioning and ground control validation. Drone Pilot: A1-A3, A2 Licenses.

Photogrammetry & point clouds: Agisoft Metashape, WebODM, ArcGIS Drone2Map, CloudCompare, DJI Terra, DJI Image Processor

GIS & remote sensing: ArcGIS Pro; Calculation and usage of vegetation indices; thermal anomaly detection; LiDAR terrain derivatives (slope, curvature, relief)

Statistics & modelling: PCA, Mann-Whitney U, Spearman correlation, Benjamini-Hochberg correction, binary logistic regression, Random Forest, stratified k-fold cross-validation, ROC and precision-recall evaluation; spatial statistics. SPSS and Python.

LANGUAGES

Greek: native · English: C2, ECPE Certificate of Proficiency (University of Michigan, 2013) · French: B2, DELF (2016)

REFEREES

Assoc. Prof. Christos Vasilakos, Department of Geography, University of the Aegean, Mytilene. chvas@aegean.gr

Assoc. Prof. Emeri Farinetti, Dipartimento di Studi Umanistici, Roma Tre University, emer.farinetti@uniroma3.it

Dr. David Scahill, Research Associate, Austrian Archaeological Institute, Athens Branch, [david.scahill\(at\)oeaw.ac.at](mailto:david.scahill(at)oeaw.ac.at)